

## NRES 746, Week 8, Lecture 1: Newton's Method by Hand (take-home)

Name: \_\_\_\_\_

**Bring this to class on Wednesday.** We'll start by going over it, and Wednesday's lecture picks up where it leaves off. Work on paper; a calculator is all you need.

### Putting it in context

Since Week 6 we've found MLEs with `optim()` without asking how it finds them. On Wednesday we open the black box. This WOD works through the core of one method, Newton's method, on a problem small enough to do by hand.

The data are the Week 7 pika haypile counts from six talus plots: 2, 5, 3, 0, 4, 4 (sum 18), modeled as Poisson with mean  $\lambda$ . Dropping terms that don't involve  $\lambda$ ,

$$\text{NLL}(\lambda) = 6\lambda - 18 \log \lambda$$

### Part 1: Slope and curvature

**1a.** Find  $\text{NLL}'(\lambda)$  and  $\text{NLL}''(\lambda)$ . Then set  $\text{NLL}'(\lambda) = 0$  and solve for the MLE,  $\hat{\lambda}$ .

## Part 2: Slopes without calculus

`optim()` can't do calculus. It estimates slopes from NLL values at nearby points, using a **finite difference**: for a small step  $h$ ,

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

Here are NLL values at three points spaced  $h = 0.5$  apart:

| $\lambda$        | 1.5    | 2.0     | 2.5     |
|------------------|--------|---------|---------|
| NLL( $\lambda$ ) | 1.7016 | -0.4766 | -1.4932 |

**2a.** Estimate  $\text{NLL}'(2)$  two ways: with the formula above (using  $\lambda = 2$  and 2.5), and with a centered version that uses  $\lambda = 1.5$  and 2.5. Compare both with the exact value from 1a. Which is closer?

Forward: \_\_\_\_\_      Centered: \_\_\_\_\_      Exact: \_\_\_\_\_

**2b.** Curvature is the slope of the slope. Use that idea to build a finite-difference approximation to  $f''(x)$  that uses only  $f(x-h)$ ,  $f(x)$ , and  $f(x+h)$ . Then use it to estimate  $\text{NLL}''(2)$  and compare with the exact value.

Estimate: \_\_\_\_\_ Exact: \_\_\_\_\_

### Part 3: Newton's method

Newton's method looks for the point where  $\text{NLL}'(\lambda) = 0$ . At the current guess, it replaces the NLL with a parabola that has the same slope and curvature, then jumps to the bottom of that parabola:

$$\lambda_{\text{new}} = \lambda - \frac{\text{NLL}'(\lambda)}{\text{NLL}''(\lambda)}$$

**3a.** In words: what does dividing by the curvature do to the size of the step? When will Newton's method take big steps, and when small ones?

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**3b.** Start at  $\lambda = 2$  and take three Newton steps, using your exact derivatives from 1a. Keep at least three decimal places.

| Step | Guess | Slope, NLL' | Curvature, NLL'' | Next guess |
|------|-------|-------------|------------------|------------|
| 1    | 2     |             |                  |            |
| 2    |       |             |                  |            |
| 3    |       |             |                  |            |

**3c.** Now start at  $\lambda = 7$  and take one step. What went wrong? Sketch the NLL curve and the parabola Newton's method used at  $\lambda = 7$  to explain why, and suggest one way to fix it.

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**3d.** (*If you have time.*) `optim()`'s derivative-based methods don't know the exact derivatives. Take one Newton step from  $\lambda = 2$  using your finite-difference estimates from 2a (centered) and 2b instead. How close is it to your first step in 3b?